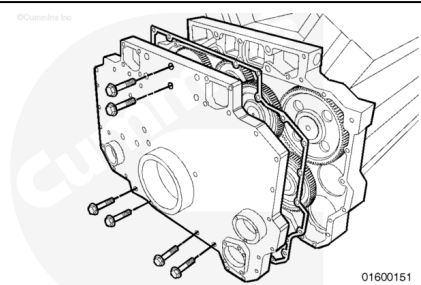


# Install

## with Mechanically Actuated Injector

This illustration shows the location for the six 16 mm capscrews. All of the other front cover capscrews are 12 mm.



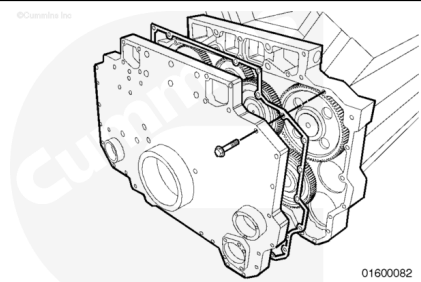
**⚠ WARNING ⚠**

**This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.**

Do **not** tighten the capscrews until the master dowel pin is installed.

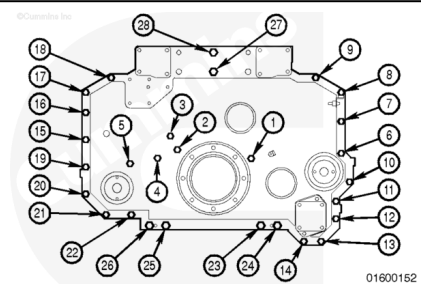
The use of guide bolts will aid the assembly.

Install the front cover gasket, cover, and capscrews.



Tighten the 28 mounting capscrews in the sequence shown.

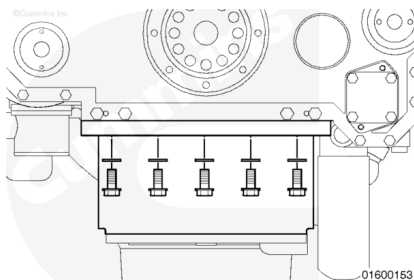
|               |         |             |
|---------------|---------|-------------|
| M12 capscrews | 80 n.m  | [59 ft-lb]  |
| M16 capscrews | 195 n.m | [144 ft-lb] |



Install the five capscrews that attach to the bottom of the oil pan adapter.

Tighten the capscrews.

**Torque Value:** 45 n•m [ 33 ft-lb ]



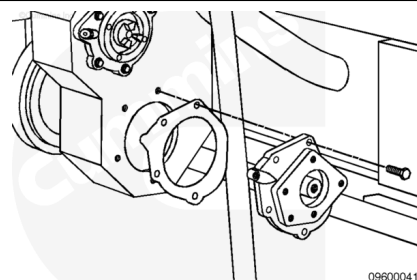
Install the hydraulic pump drive.

Use clean engine oil to lubricate the bushing in the front gear cover.

Install the gasket, drive or cover, washers, and capscrews.

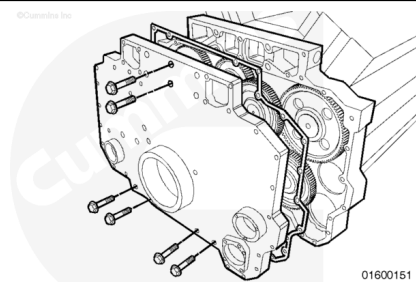
Tighten the capscrews.

**Torque Value:** 45 n·m [ 33 ft-lb ]



## with Electronically Actuated Injector

This illustration shows the location for the six 16 mm capscrews. All of the other front cover capscrews are 12 mm.



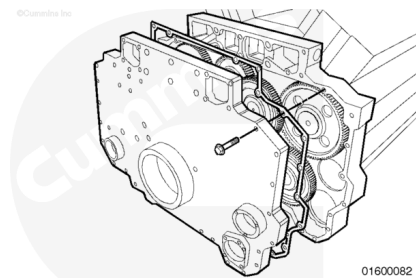
### **⚠ WARNING ⚠**

**This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.**

Do **not** tighten the capscrews until the master dowel pin is installed.

The use of guide bolts will aid the assembly.

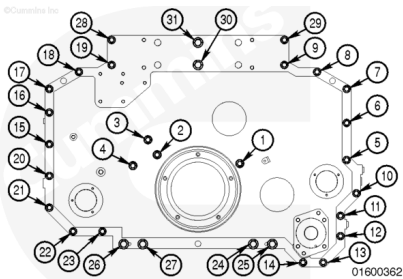
Install the front cover gasket, cover, and capscrews.



**Note :** The number of capscrews that can be tightened at this time depends on the engine options. If the location is used to mount an accessory, it can **not** be installed at this time.

Tighten the front gear cover mounting capscrews in the sequence shown.

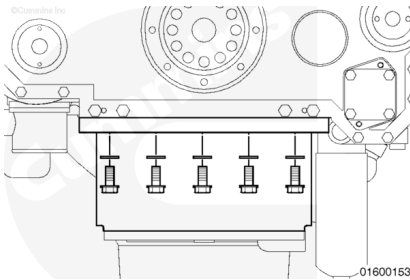
|               |         |             |
|---------------|---------|-------------|
| M12 capscrews | 80 n.m  | [59 ft-lb]  |
| M16 capscrews | 195 n.m | [144 ft-lb] |



Install the five capscrews that attach to the bottom of the oil pan adapter.

Tighten the capscrews.

**Torque Value:** 45 n•m [ 33 ft-lb ]



Install the hydraulic pump drive.

Use clean engine oil to lubricate the bushing in the front gear cover.

Install the gasket, drive or cover, washers, and capscrews.

Tighten the capscrews.

**Torque Value:** 45 n•m [ 33 ft-lb ]

